

input pair: $\mathcal{N}_1, \mathcal{N}_2$

is an instance of

pair of join-subnetworks:

$$\begin{aligned}\mathcal{N}_1 &= \mathcal{N}_1^1 * \dots * \mathcal{N}_1^p \\ \mathcal{N}_2 &= \mathcal{N}_2^1 * \dots * \mathcal{N}_2^q\end{aligned}$$

Rules 1&2

matching pair of join-subnetworks:

$$\begin{aligned}\mathcal{N}_1 &= \mathcal{N}_1^1 * \dots * \mathcal{N}_1^p \\ \mathcal{N}_2 &= \mathcal{N}_2^1 * \dots * \mathcal{N}_2^p \\ \forall i \in [p] \quad L(\mathcal{N}_1^i) &= L(\mathcal{N}_2^i)\end{aligned}$$

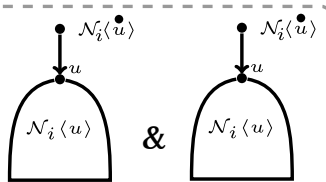
f_C

recursive equations

lower nodes

decomposition

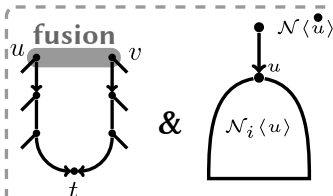
pair of prime networks with same leaf-set:



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&

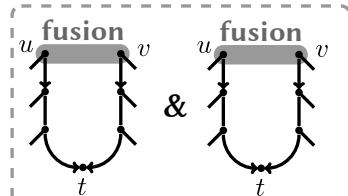
or



&

&

or



f_P

n-subnetworks:

$$\cdot * \mathcal{N}_1^p$$

$$\cdot * \mathcal{N}_2^p$$

$$= L(\mathcal{N}_2^i)$$

$$f_C$$

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